activation 200 kJ mol^{-1}. If the frequency factor of the reaction is $1.6 \times 10^{13}\ s^{-1}$. Calculate the rate constant at 600 K. $\left(e^{-40.09} = 3.8 \times 10^{-18}\right)$

20. For the reaction $2x + y \longrightarrow L$ find the rate law from the following data.

[x] (min)	[y] (min)	rate ($M\ s^{-1}$)
0.2	0.02	0.15
0.4	0.02	0.30
0.4	0.08	1.20

21. How do concentrations of the reactant influence the rate of reaction?
22. How do nature of the reactant influence rate of reaction.
23. The rate constant for a first order reaction is 1.54x 10-3 s-1. Calculate its half life time.
24. The half life of the homogeneous gaseous reaction $SO_2Cl_2 \rightarrow SO_2 + Cl_2$ which obeys first order kinetics is 8.0 minutes. How long will it take for the concentration of SO_2Cl_2 to be reduced to 1% of the initial value?
25. The time for half change in a first order decomposition of a substance A is 60 seconds. Calculate the rate constant. How much of A will be left after 180 seconds?
26. A zero order reaction is 20% complete in 20 minutes. Calculate the value of the rate constant. In what time will the reaction be 80% complete?
27. The activation energy of a reaction is 225 k Cal mol^{-1} and the value of rate constant at 40°C is $1.8 \times 10^{-5}\ s^{-1}$. Calculate the frequency factor, A.
28. Benzene diazonium chloride in aqueous solution decomposes according to the equation $C_6H_5N_2Cl \longrightarrow C_6H_5Cl + N_2$. Starting with an initial concentration of 10 g L^{-1}, the volume of N_2 gas obtained at 50 °C at different intervals of time was found to be as under:

t (min):	6	12	18	24	30	∞
Vol. of N_2 (ml):	19.3	32.6	41.3	46.5	50.4	58.3

Show that the above reaction follows the first order kinetics. What is the value of the rate constant?

29. From the following data, show that the decomposition of hydrogen peroxide is a reaction of the first order:

t (min)	0	10	20
V (ml)	46.1	29.8	19.3

Where *t* is the time in minutes and V is the volume of standard $KMnO_4$ solution required for titrating the same volume of the reaction mixture.

30. A first order reaction is 40% complete in 50 minutes. Calculate the value of the rate constant. In what time will the reaction be 80% complete?

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